

Chapter 177

Mixing in the Fairlight Page

The Mixer is “control central” of the Fairlight page, providing all of the functionality you need to mix the various audio tracks of your program into a harmonious whole.

By using EQ, dynamics, panning, level control, and Fairlight FX, VST and Audio Units audio plugin effects of all kinds, all with full automation, you can hone your sound and balance each track’s elements with one another.

This chapter focuses on explaining the various functions of the Mixer so you can harness its power for yourself.

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Introduction to Mixing

This chapter describes how to use the Mixer to adjust the levels and fine-tune the audio of each track in the timeline. It's focused on the function of the channel strip controls, with the following exceptions:

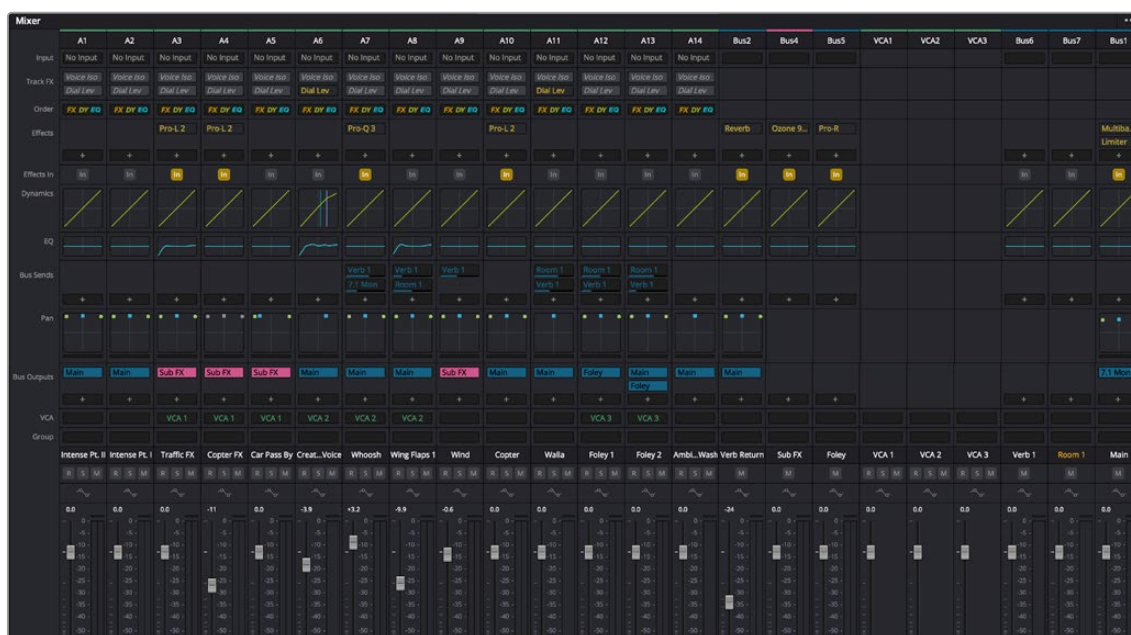
For more information about bussing, see *Chapter 171, "Setting Up Tracks, Busses, and Patching."*

For more information about recording audio, see *Chapter 173, "Recording."*

For more information about mix automation, see *Chapter 178, "Mix Automation."*

The Mixer

The Audio Mixer provides a set of graphical controls you can use to assign track channels to output channels, adjust EQ and dynamics, set levels and record automation, pan stereo and surround audio, mute and solo tracks, and add Fairlight FX, or VST, or AU effects plugins. At its most basic, each audio track in your timeline corresponds to an individual channel strip in the Mixer, and by default there's a single main stereo mix bus labeled "Bus 1" that combines all these tracks into an overall mix.



The Audio Mixer with channel strips corresponding to the tracks in the Timeline

Tracks and Busses

Once you start creating busses, the Audio Mixer exposes two sets of channel strips. The leftmost set of channel strips correspond to the audio tracks in the Timeline, while the right-most set of channel strips expose sets of controls for each bus that you've created.



By default, the Audio Mixer is divided into two sections, one for tracks (at left) and one for busses (at right).

If you have more tracks and busses than can be displayed all at once given the width of your computer display, then each half of the Mixer has independent scroll bars so you can choose which tracks and which busses you want to see next to one another. You can also remove the split between sections and have a single, continuously scrollable view with all tracks, or change the order or split point for the channels.

FlexBus Routing and Mixing

The Fairlight audio engine bussing system is called FlexBus. As its name suggests, FlexBus offers a high level of flexibility for bus types and signal routing. DaVinci Resolve Fairlight's older Fixed Bus configuration (prior to DaVinci Resolve v17) offered a more limited fixed bus topology, and can still be used, but FlexBus makes it possible to patch outputs and/or sends in any way required, as dictated by your project.

With FlexBus, each track can output to up to ten busses and sends with additional level and pan controls to a further ten busses. Busses can be sent to other busses up to six layers deep, facilitating complex stem building, processing, and allowing discrete deliverables.

For more information, see *Chapter 171, "Setting Up Tracks, Busses, and Patching."*

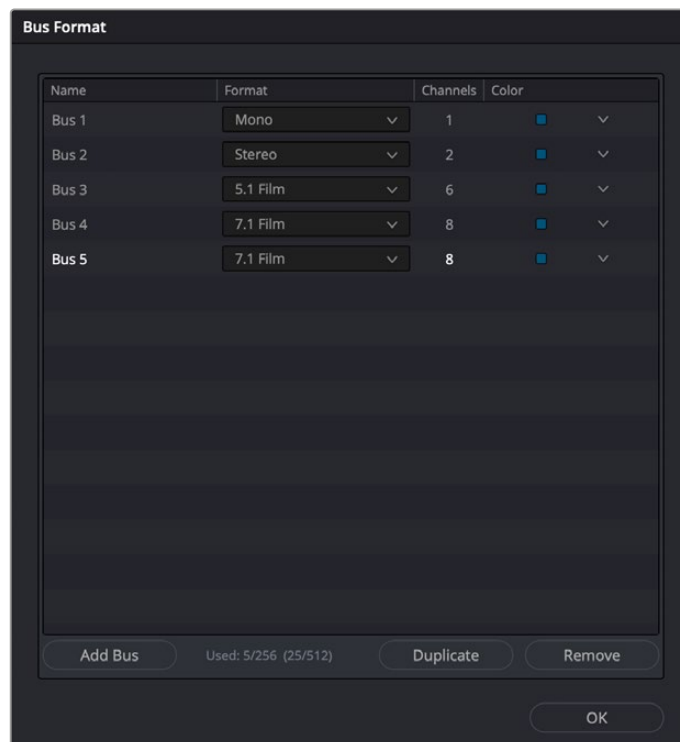
What is a Bus?

A bus is simply a common signal connection point in an audio mixer. Busses can be mono, stereo, or any larger format, like 5.1 or Dolby Atmos 9.1.6 (where 16 audio signals are used). Bussed connections are mix points for the audio, allowing each audio track feeding their signals to the bus to be summed together to create a master output for that bus.

Bus-to-Bus Routing

A mix may use a single bus (for example, the default stereo Main 1 bus may be all that you need if you're doing basic stereo mixing). Or you can create "submixes" where multiple busses feed into other busses to create a master mix. For example, you might have a dialogue bus, an effects bus, and a music bus so you have control over those elements separately via the bus "masters" for each, then feed the 3 masters into a main output bus for final mix rendering.

You may want to route the output of a bus to an audio track to record the resulting signal, or route to monitor speakers to listen to the real time output. Finally, the Deliver page can render the output of your designated master bus to create a final bounced file.



The FlexBus structure allows for many different bus track types to be created or changed.

User-definable busses allow for bus-to-bus, bus-to-track, or track-to-bus routing, with each bus having the ability to sum together signals from mono to fully immersive formats (such as Dolby Atmos and Ambisonics) at your discretion. As with any and all of the audio track types in Fairlight, you can change bus types at any time if needed.

The power of the FlexBus system is that it allows you to direct signals to many different places at one time to achieve complex mixing scenarios. Perhaps you need to generate two mixes that are identical in content but need to be of different output levels. You can designate two mix busses, one with a target output level of -14 LUFS and one with a target output level of -24 LUFS. The final mix signal is sent to one bus that is then broken into two more busses, one with a limiter set to accommodate the -14 LUFS mix and one set to the correct level for the -24 LUFS mix, allowing you to create these two different mixes at one time.

In Ambisonic workflows, you can route an Ambisonic bus to another Ambisonic bus, even with differing orders, and any required up- or down-mixing is automatically handled by the Fairlight engine and FlexBus.

FlexBus supports expanded Dolby Atmos capabilities and Ambisonics; for more information see *Chapter 184, "Immersive Audio Workflows."*



FlexBus bus routing buttons

Legacy Fixed Bussing

Prior to DaVinci Resolve 17, the Mixer had a fixed topology with specific, fixed numbers of sends, submix busses, and main outputs. DaVinci Resolve still supports this older “legacy” bussing topology to maintain compatibility with earlier projects.

With the current FlexBus bussing, the Mixer looks slightly different. The Aux panel on the legacy Mixer has been renamed Bus Send, and the Main/Submix panel has been changed to Bus Outputs.

In the legacy Fixed Bus view:

- Each audio track’s channel strip has a set of Main and Submix bus buttons that let you assign the audio output by that channel strip to a Sub (typically used to combine different subsets of tracks into submixes) or to a Main (typically used to output or render overall mixes).
- Each Sub and Aux’s channel strip has a set of Main buttons so that different combinations of Subs can be assigned to each Main. Main channel strips have no buttons because, from a bussing perspective, they’re the final output.
- Main and Submix buttons let you assign tracks to busses, and busses to other busses.



Main and Submix bus buttons

Using Legacy Fixed Busing

If you want to work using the previous method of Fixed Bus mapping, you can do so for new projects by accessing the Project Settings > Fairlight panel, and turning on the “Use fixed bus mapping” checkbox. However, as there is no practical advantage to using the older legacy fixed bus system, and several features are only available in FlexBus, using FlexBus is recommended.

Converting Older Fixed Bus Projects to FlexBus

Older Fixed Bus projects can be converted to FlexBus by doing the following:

- Open Project Settings > Fairlight.
- Under the Bussing heading, uncheck “Use fixed bus mapping.”
- A dialog will appear, allowing you to convert the project to FlexBus.

Customizing the Onscreen Mixer Controls

The 3 dot Option menu at the upper right-hand corner of the Mixer provides several different options for customizing the appearance of the Mixer, as well as which controls are shown or hidden:

- **Full Track Mixer or Small Track Mixer option:** While the Full Track Mixer provides more room for buttons and controls that are large enough you don't have to squint to see them, the Small Track Mixer allows many more channel strips to be seen at one time.



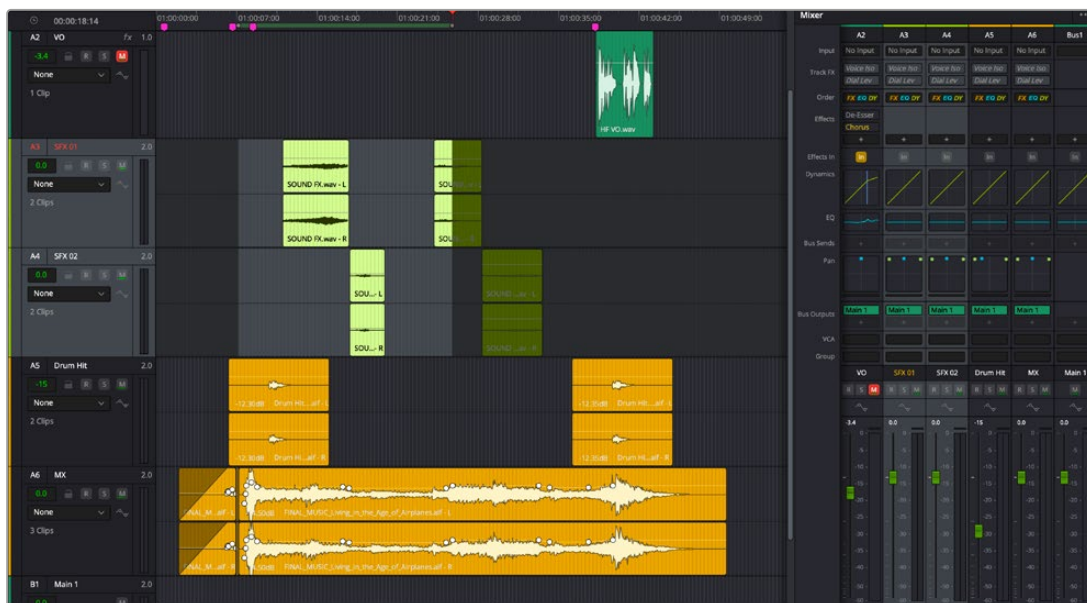
The Small Track Mixer view

- **Show Labels option:** These are the sectional names for functional areas of the Mixer that appear at the far left of the Mixer channel strip area. If you're first getting started, these labels are useful, but once you've gotten used to the Mixer's layout, hiding the labels can give you a bit more room to work with for another channel strip or two.
- **Channel Strip Control Visibility options:** Allows controls to be shown or hidden in the Mixer. You can hide controls you know you're not going to be using, no longer need to adjust, or wish to not disturb.

Selecting Channel Strips and Tracks

When you select a track in the Timeline, you also select that track's accompanying Mixer channel strip, and vice versa:

- To select single tracks, just click.
- To select multiple contiguous tracks, select a starting track or channel strip, hold down Shift, and click on last item you want in your selection.
- You can also select multiple channel strips in the Mixer by clicking and dragging to create a bounding box around the tracks you want to select.
- To select non-adjacent (discontiguous) tracks or channel strips, command-click on each item you want to add to your selection.

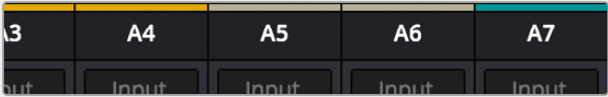


Selecting multiple channel strips selects the same tracks in the Timeline

Track Organization

Each channel strip has three organizational properties that let you keep track of which channel strip is responsible for which part of the mix.

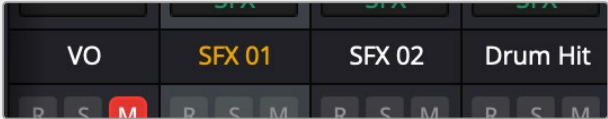
Track color: Each track can be differently color coded to help you keep organized. These colors appear in the Timeline track header, the Mixer, and the meters to help you keep track of which track corresponds to which channel strip and meter.



The track color and track number appear at the top of each channel strip.

Track number: The number of the timeline audio track corresponding to each channel strip appears here.

Track name: This mirrors the track name found in the track header of a track on the Timeline. If you customize an audio track's name in the Timeline or in the Mixer, that name appears here.



Each track's name appears between the assignment buttons and the arm/solo/mute buttons

Rearranging Tracks

You can reposition your tracks and busses in the Mixer by clicking and holding the track name and dragging left or right to the desired location. This works for both individual and selection of multiple channel strips.

As you move your mouse, a white line will appear indicating the new location and the change will be reflected in the Index.



Rearranging tracks by moving a Mixer channel strip

Disabling Tracks

You can disable an audio track in the Mixer by right-clicking the track number at the top of the corresponding channel strip and choosing **Disable Track** from the contextual menu. Repeating these steps will re-enable the track.

To simultaneously disable multiple tracks, select them, Option right-click any track number in the group, and choose **Disable Tracks** from the contextual menu.



Disabling multiple tracks via the Mixer

Managing Channel Strips Using the Index

The Index is a consolidated list view of all the tracks in the current Timeline, designed to make it easier to manage timelines with lots of tracks. Columns let you see each track’s visibility, track number, name, controls, channel format, monitor assignment (if any), ADC controls, and VCA assignment. The controls in these columns let you manage the tracks in the current Timeline by showing or hiding them, toggling track controls, or rearranging them. You can also right-click on one or more selected tracks to color code them, or change the track type.

Edit Index Tracks Markers									
Tracklist									Q
	#	Name	Track Controls				Format	VCA	
👁	A1	SYNC	🔒	R	S	M	1.0	VCA 1	
👁	A2	VO	🔒	R	S	M	1.0	VCA 1	
👁	A3	SFX 01	🔒	R	S	M	2.0	VCA 2	
👁	A4	SFX 02	🔒	R	S	M	2.0	VCA 2	
👁	A5	Drum Hit	🔒	R	S	M	2.0		
👁	A6	MX	🔒	R	S	M	2.0		

The Tracks panel of the Index shows all tracks on both the Timeline and the Mixer.

Showing and Hiding Tracks

You can use the Tracks panel of the Index to hide tracks you don’t need to work on in order to create more room for tracks you need to see.

TIP: To quickly show or hide a number of tracks, click and drag up or down over the eyeball button of each track where you want to toggle the visibility.

Color Coding Tracks

Select one or more tracks, then right click one of the selected tracks and choose a color from the Change Track Color submenu of the contextual menu.

Toggling Lock, Record, Solo, and Mute

As with visibility, you can use the Lock, Record, Solo, and Mute controls to quickly enable or disable multiple tracks by clicking and dragging up or down over the Lock, R, S, or M buttons you want to toggle.

Rearranging Tracks

You can rearrange tracks in the Tracks tab in the Index by clicking any track strip area between other controls, and then dragging that track up or down in the Index. As you drag, a white line shows you where that track will be inserted when you release it. You can even select a contiguous series of tracks and drag them all to a new position in the Timeline at once.

Managing the Dividing Line

By default, there are two areas of tracks in the Index; one area is for audio tracks, and one area for busses and VCAs with a dividing line separating them. Busses and VCAs can be dragged up out of the divider and into any position for any order. This can be quite useful when having a series of dialogue tracks, for instance, and pulling the dialogue bus up next to them.

The dividing line's position can be moved to any location by dragging it, so you can keep the split organization of tracks into two different areas but re-order what is shown.

Once the order has been changed in the Index, this is reflected in both the Mixer panel as well as in the Meter panel. By reordering busses and tracks, you can adjust your workflow for whatever set of tasks you are working on.

Single Mixer View

Clicking the three-dot Option menu at the top of the Mixer panel offers an option for Single Mixer view. The Single Mixer view removes the divider line from the Tracks tab in the Index entirely, providing a continuous scroll of the tracks, both in the Mixer panel horizontal scroll and the Index vertical scroll. You can restore the divider at any time.

NOTE: You can reset the bus order to its original layout by choosing “Reset Bus Order” from the three-dot Options menu on the upper right of the Mixer.

Changing Track Type

Track type refers to the channel width of the audio track and what format it is. The track type can be changed at any time. For example, you could have a mono track and change it to a stereo, or 5.1 track, or a 12 channel adaptive.

To change the track type, select one or more tracks, then right-click one of the selected tracks and choose a different track type from the Change Track Type submenu of the contextual menu.

NOTE: Keep in mind that if you change from a larger track type to a smaller one, you may need to also set which of the channels in the larger source audio file are now audible on the smaller track type. This can be handled via the contextual menu choice Clip Attributes for a clip, using the Audio tab.

Deleting Tracks

You can delete an individual track by right-clicking it and choosing Delete Track from the contextual menu.

To delete multiple selected tracks simultaneously, right-click any track within the collection and choose Delete Tracks from the contextual menu.

Fairlight Mixer Signal Path

The signal path or layout (“topology”) of a mixer describes how the audio signals are routed from one place to another.

The default signal flow on the Fairlight channel strip is as follows:

- 1 Audio source: Input menu allows choice of source, including file playback, bus, live input, etc.
- 2 Path Settings: Accessible from the Input menu. Allows adjustments to the input signal.
- 3 Track FX: Built-in effects, Voice Isolation (Studio only), and Dialogue Leveler. Disabled by default.
- 4 Effect: Fairlight FX, AU, or VST effects.
- 5 Dynamics: Dedicated channel dynamics processor with expander/gate, compressor, and limiter.
- 6 EQ: Dedicated 6 band EQ with variable gain, frequency, and Q.
- 7 Output Routing: Panning and bussing.

The main fader for the channel strip then controls the level to any assigned main output busses. Bus Send faders control the level to any destination set for a mixer channel send and can be set pre-or post-fader.

What Is An Effects Insert?

Each time you add an effect to a channel, a signal processing “insert” (insertion point) is enabled. The source signal flows from the input of the effect to its output, and then onwards to any additional insert(s) to the next stop in the Mixer’s signal flow.

DaVinci Resolve supports a total of six inserts per channel in addition to the built in Dynamics and EQ processing.

Click on the Input area to access the Input dropdown menu, letting you patch inputs, patch busses, and configure the input settings of audio signals routed through Blackmagic or other hardware audio interfaces.

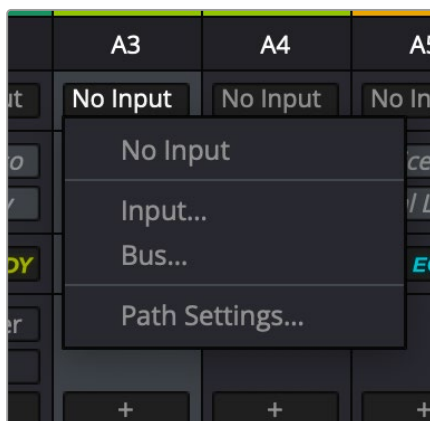
“Pre” vs. “Post”

The term pre refers to a signal that is routed for use before another connection point. For example, you may prefer to work with pre-fader metering as you can always see the original signal level coming off of disk reflected in the meter level.

A signal that is post occurs after a connection point. For example, you generally want effect sends that go to a reverb effect to follow the level of a track’s main fader so that the reverb level follows the overall level. As a result, Bus Sends are set to have post fader routing by default.

Input

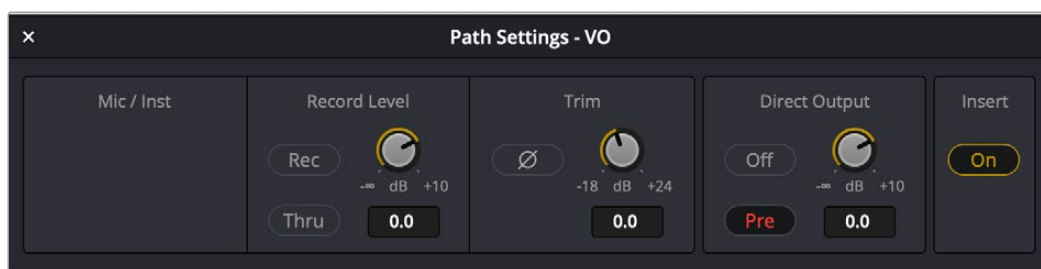
For more information on using the options of the Input menu, see *Chapter 171, “Setting Up Tracks, Busses, and Patching.”*



The Input dropdown menu allows you to initiate patching any available sources using the Patch Input/Output window.

Path Settings

Choose Input at the top of the Mixer channel strip to access the Path Settings control window. These controls allow you to modify various aspects of the Mixer channel strip that may need adjustment prior to, or post, recording to control the mic/line inputs on Fairlight audio interface hardware, and more.



Path Settings window

Mic/Inst

Controls will only appear on this panel if you have connected channels 1 or 2 of a Fairlight SX36 audio interface to your system. If connected, you can remotely control all of the options (including level) for the mic/instrument inputs of the SX36 if they are assigned to the channel. If there is no connection to an SX36, the area is empty.

Record Level

- **Record:** Linked and identical to the Record Enable button on the channel strip; here for convenience. If you hit one, it will enable the other.
- **Thru:** Allows the input signal to pass into the Mixer without enabling a record path. This is ideal when you want a source signal to always be available and just want to monitor it.
- **Record Level:** Allows you to apply a digital gain adjustment to the record path to disk, post the output of your audio interface's analog-to-digital converter.

Normally, this control should be left at 0.0 (no change, unity gain), as it affects the level you are recording to disk. It is best practice to use the level controls on your audio interface to control the input level into DaVinci Resolve in order to maximize audio fidelity. However, there might be a time where you need a bit more level, or may not have access to an audio interface's controls, and in those cases you can adjust the input.

Trim

- **Polarity:** Inverts the polarity of the signal coming into the channel strip (sometimes referred to as “flipping the phase”). For example, you may have an input signal like an explosion where the transient attack of the signal produces with a massively positive-going waveform (where the waveform mostly appears above the zero line). If you invert the polarity, the signal will now be mainly negative-going and the waveform will be concentrated beneath the zero line. Inverting polarity is sometimes used to more closely align signals from multiple microphones and can be used creatively to affect frequency response of such a signal.
- **Trim Level:** Allows you to trim the level of the signal coming off of disk, post the Track FX (which process directly off of disk) and pre all other effects. Trim Level appears post the recorded signal to disk and doesn't affect the recording level. Trim is a playback adjustment and allows you to adjust the level of the signal coming into your mix to optimize the level feeding effects and the busses, or to slightly trim an otherwise perfect element in the mix up or down in level.

Direct Output

Each audio channel strip can enable a direct output that can be used to feed any other input destination. You can patch this source using the Track Direct choice in the Patch Input/Output dialog via the Source dropdown.

For more information see *Chapter 171, “Setting Up Tracks, Busses, and Patching.”*

- **On/Off:** Enables/disables the direct output.
- **Pre:** Sets the direct output tap-off point to be pre (before) the channel fader. On by default.
- **Level Control:** Sets the level from full off (minus infinity) to +10 dB. Default is 0.0 (unity gain).

Insert

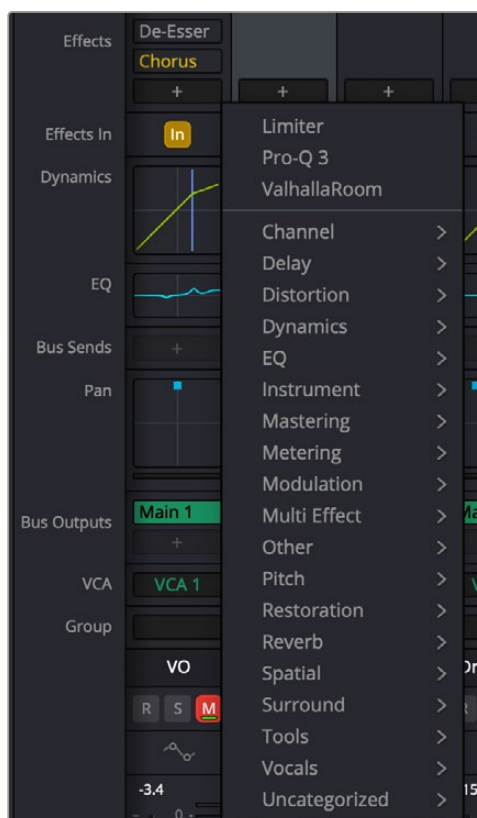
Linked and identical to the Effects In button on the channel strip; here for convenience. If you hit one, it will enable the other. Switches all Fairlight FX, AU, or VST effects on a channel in or out of the signal path with a single switched control.

Effects

Fairlight FX, VST, or Audio Unit effects can be applied on the Mixer's channel strips. Effects are applied on the entire track, not just on one or more clips in the Timeline for that track.

Clicking the plus (“+”) button opens a dropdown menu, allowing you to apply any Fairlight FX, VST, or AU effects installed on your machine to that track. The menu is organized by processing type. Within each type, there is a hierarchical choice for Fairlight FX, AU, and VST (if you're on a Mac) or Fairlight FX and VST (if you're on Windows).

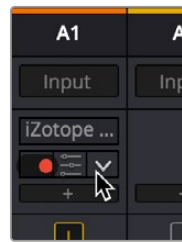
NOTE: Depending on what you have installed on your machine, the collection of audio effects available for the Mixer channel strip inserts is often close to (or identical to) those that you can access for clip-based audio processing found in the Effects panel on the left side of Cut, Edit, and Fairlight pages.



Clicking on the plus sign shows the Effects dropdown list for assignment

TIP: You can store favorite effects for quick access at the top of the list by marking them as favorites in the Effects panel. Mark a favorite by clicking on the star next to the name.

Hovering the mouse over any listed effect reveals controls for enabling/disabling, replacing or deleting the effect, or opening an effect's custom controls (the user interface for that effect).



Hovering the mouse pointer over an effect insert reveals controls for that effect.

Changing the Order of Effects on a Channel

The order of effects on any Mixer channel strip can be changed by clicking on the Effects Order button. A dropdown menu appears allowing you to change the order between various routing combinations of all insert effects (Fairlight FX, or AU, or VST plugins) and the built-in Dynamics and EQ modules. The default routing is Effects ("FX"), Dynamics, EQ.



Effects Order dropdown menu

Different effects orders can produce very different creative results. For example, it is usually best practice to apply noise reduction first in a signal chain, so you're not affecting the volume or quality of the noise floor you're trying to remove, so the plugin can do its job most effectively. Or you may wish to apply dynamics processing prior to EQ, as changing the spectral content of your source may affect the response of the compressor.

Moving Inserts by Dragging

You can move inserts from one location to another simply by clicking and dragging to a new location. Using this technique, you can move an effect from one channel to another, or swap or move the location of an effect on a channel to another insert position on that channel (displacing the plugin already present). Note that, at this time, you can't use this technique to copy tracks to busses or vice versa.

Copying and Pasting Effects

You can copy entire effect chains with their specific effect parameters intact, track by track. Let's say that you've created a plugin chain on a dialogue track that has Noise Reduction, a De-Hummer, and a Dynamics plugin that you have tweaked to the exact settings needed for those dialogue recordings to sound great.

You also have another track that has similar recordings that could benefit from the exact same chain of plugins and those particular settings. Rather than install each plugin to the track and redo the settings, you can simply right-click the track header (A1 for example) and click Copy. Then go to the track header of the new track you want effected, right-click, and click Paste. You will now have the identical plugin chain copied over to the new track, all with each of the settings exactly as you had created on the source plugin dialogue track

Copying and Pasting Effect Settings

If you have a setting for one effect that you spent a lot of time getting right and see that it could also work just as well on another track's effect, you can copy and paste just those settings.

For example, you have a Reverb on a dialogue track that you have tweaked to perfection. You see that those exact settings will work well for another dialogue track that is on your timeline. Rather than try to recreate those settings, you can duplicate them.

- Hold down the option key and drag the Reverb plugin you want to copy to a new location. This creates a copy at the new location and leaves the original intact.
- Alternatively, you can use Copy and Paste in the plugin three dot Options menu. To apply those settings to the other track, just click Copy from the Options menu of the source Reverb effect, then open the Reverb plugin window of the destination Reverb effect, and click the Options menu to paste. Now all of the parameters are the identical between the two plugins.

For more information on using audio effects, see *Chapter 180, "Audio Effects."*

TIP: You can create and load presets for entire channel strips and use them in any project or timeline using the Fairlight > Presets Library.

Effects In

The Effects In button allows you to bypass all Fairlight FX, AU, or VST effects plugins for the channel with a single control. When it is lit (orange), all effects are enabled; when it is unlit, all effects are bypassed.

NOTE: Track FX (Voice Isolation with Resolve Studio and Dialogue Leveler) and the built-in Dynamics and EQ processors are not affected by Effects In.

Dynamics

Each Mixer channel strip has a built-in Dynamics processor, with a mini Dynamics Indicator graph that appears on the channel strip and acts as the access button. Double-clicking the Dynamics Indicator brings up the Dynamics processor window, with three modules: an Expander/Gate, a Compressor, and a Limiter. Each can be used separately or in concert to manage the dynamics of the audio on the target track.

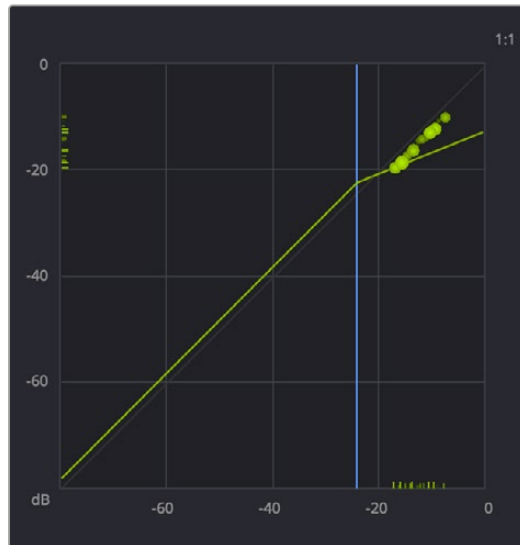


The Dynamics control window

The Dynamics Graph

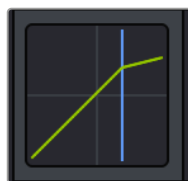
When you enable any combination of the Dynamics window modules you may need for a track, the Dynamics graph updates with curves that show you how the signal is being affected by the Expander/Gate, Compressor, and/or Limiter settings that are being applied. The Dynamics graph also provides a real time animated display that meters the true response of the input/output signals compared to the graphic that shows the basic response curve.

- The X axis (horizontal scale) represents the input signal level.
- The Y axis (vertical scale) represents the output level.
- The blue line represents the threshold.
- The light green line represents the dynamics curve (dependent on the process type and setting; this will change).
- The light green balls that animate in the upper area of the graph follow how the Dynamics control settings affect the signal as it moves above or below the threshold. The balls are affected by the attack and release settings, allowing you to visualize the way the compressor is affecting the input signal.
- The smaller lines that appear on the on the input and outline X-Y lines allow you to see exactly where the input/output levels are.



The Dynamics graph shows the response curve and how the signal is being affected by the current settings in real time.

The Dynamics Indicator on the channel strip displays the same curve, so you can see at a glance what is happening on that channel (but without the real time display). This indicator cannot be adjusted; you must open the Dynamics window to make modifications.



The channel strip
Dynamics Indicator

Master Dynamics Controls

The Dynamics window contains the following overall controls at the top of the window:

- **Enable button:** Turns the overall dynamics effect off and on, without resetting the controls.

NOTE: You can also turn dynamics on/off by single clicking on the Dynamics button (the mini graph) in the channel strip.

- **Reset button:** The circular reset icon on upper right next to the three dot Options menu resets all controls of the Dynamics window to their defaults.
- **Preset Menu:** Dynamics presets appear here. You can create, change, and save presets, and use the default presets as good starting points for your track's specific needs.
- **Make Up Gain control:** A post-dynamics level control that lets you boost the signal to compensate for dynamics settings that may have lowered the overall level.

Expander/Gate

The first of three sets of dynamics modules, the Expander/Gate controls can be switched between expansion and gating. Expansion emphasizes differences in volume by lowering the level of soft parts of the signal relative to the level of louder parts and can be used to minimize noise while increasing the dynamic range of a signal. Gating reduces the level or even silences parts of a signal that fall below the set threshold level in order to reduce or eliminate noise in quiet parts of a recording.

- **Expander:** This button enables the Expander.

Threshold: Sets the signal level below which expansion occurs. Defaults to -35 dB. The range is from -50 to 0 dB.

Range: Amount of decrease in signal level in dB, affected by both threshold and ratio. As ratio increases, range can have more of an effect. At lower ratios, as threshold increases, the effect of range increases.

Ratio: Sets the attenuation ratio (input to output) applied to signals which fall below the threshold level. It controls the rate at which signals will drop, while Range controls how much the signal will drop. Defaults to 1:1.1. The range is 1:1.1 to 1:3.0.

Attack: Adjusts how quickly the expansion occurs when a signal exceeds the threshold. Defaults to 1.4 ms (milliseconds). The range is 0 to 100 ms.

Hold: Controls how long the Expander is kept open after the input has fallen below the threshold, in ms. Defaults to 0 ms. The range is from 0 to 4000 ms (4 seconds).

Release: Adjusts how quickly or gradually the Expander attenuates when the input signal goes back below the threshold. Defaults to 93 ms. The range is 50 to 4000 ms (4 seconds).

- **Gate:** This button enables the Gate.

Threshold: Sets the signal level below which gain reduction occurs. The range is from -50 to 0 dB.

Hold: Controls how long time the Gate is kept open after the input has fallen below the threshold, in ms. Defaults to 0 ms. The range is from 0 to 4000 ms (4 seconds).

Range: Sets the maximum amount of gain reduction that will be applied when the signal falls below the gate threshold. Once the signal has fallen below the level determined by the gate threshold minus the gate range, no gain reduction is applied. The default is 18 dB. The range is from 0 to 60.2 dB.

Ratio: Unused for gate.

Attack: Adjusts how quickly the gating occurs when a signal exceeds the threshold. Defaults to 1.4 ms. The range is 0 to 100 ms.

Hold: Keeps dynamics from being triggered again until a certain amount of time has passed, in ms (milliseconds). Defaults to 0 ms. The range is from 0 to 4000 ms (4 seconds).

Release: Adjusts how quickly or gradually the sidechain detector stops attenuating when the input signal goes back below the threshold. Defaults to 93 ms. The range is 50 to 4000 ms (4 seconds).

Compressor

The second set of dynamics parameters let you apply compression, which detects the envelope of an audio signal in order to automatically change its level. A compressor is used to compress a signal's dynamic range by reducing differences in level between the loudest and quietest parts of the input signal.

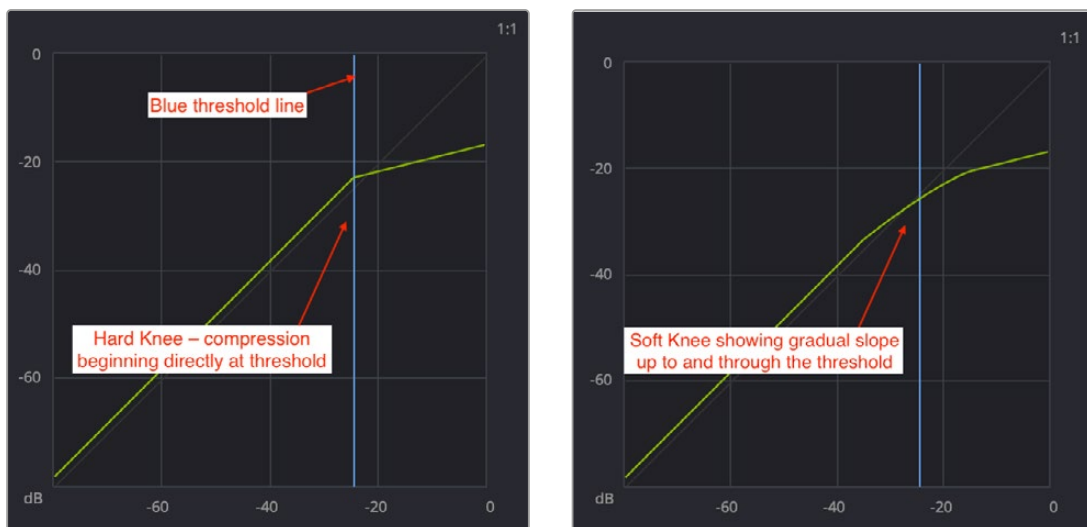
Compression can be used to raise the signal's overall level to be boosted without clipping, increasing perceived loudness. Compression is often used to allow voices to have more presence within a mix, and to smooth out changes in the levels of tracks with too much dynamic range for the task at hand.

However, “overcompressing” can remove natural dynamics of a sound, or remove too many of the natural transients (energy bursts or “attacks”) that give the original sound its character. But more extreme compression can also be used creatively to change a sound’s character to give it more perceived attack, change its release characteristic, or change the balance and blend of the environment a sound is recorded in (for example, compressing a source signal with a higher ratio and fast release to raise up the presence of a room sound that the source is recorded in).

- **Compressor:** Enables the Compressor.
- **Threshold:** Sets the signal level above which compression is applied. The default is -15 dB. The range is -50 to 0 dB.
- **Ratio:** Adjusts the compression ratio. This sets the gain reduction ratio (input to output) applied to signals which rise above the threshold level. The default is $2.0:1$. The range is $1.0:1$ to $10:1$. Commonly used ratios for dialogue processing are $2.5:1$ to $3:1$. With a $3:1$ ratio, for every 3 dB of increase in signal above the threshold, 1 dB is output.
- **Knee:** Affects behavior with signals that are very near the threshold. The knee can smooth out the transition between where you don’t hear any compression and when the “hard compression” (the set ratio) starts.

The Knee: Control allows a smoothing of the slope of the threshold point. You can see Knee’s effect on the graph when using a ratio like $4:1$ and adjusting Knee; the graph will show the line where compression “kicks in” being rounded off by higher values of the Knee control.

Normally, compression begins directly above the threshold. This is known as a “hard knee.” For a smoother and less audible handling of compression, you can set a softer knee, so the compression turns on gradually as signals approach the threshold and rise above.



Hard and Soft Knee graphs

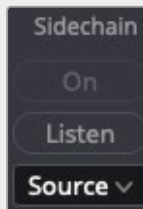
Mix: Adjusts the mix between the compressed (0) and non-compressed input signal (100), allowing you to balance the compressed signal against the original, unaffected sound. For example, if you’re creating a large explosion impact sound effect, it might sometimes be useful to mix in just a bit of the original, uncompressed signal with a more compressed version to preserve more of the original character along with compressed version. The default is 0 (all compression, no original signal).

Attack: Adjusts how quickly the compression will occur when a signal exceeds the threshold. The default is 1.4 ms. The range is 0 to 100 ms.

Hold: After the attack phase has been completed, the Hold parameter controls how long this initial attenuation is maintained, before entering the release phase. Defaults to 0 ms. The range is from 0 to 4000 ms.

Release: Adjusts how quickly or gradually the sidechain detector stops applying compression when a signal falls back below the threshold. The default is 93 ms. The range is 50 to 4000 ms (4 seconds).

Sidechain: This redesigned section lets you set up sidechain compression or “ducking” using the parameters and controls listed below, instead of using a Send from a track or bus to feed the sidechain input:



Sidechain module

On: When clicked, this button turns red indicating that the Sidechain module is active.

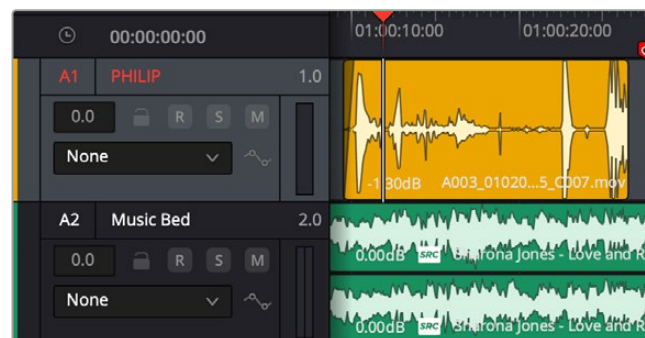
Listen: Clicking this button lets you hear the incoming signal on its own, which is helpful if you need to confirm that the correct audio signal is being sent to the Sidechain module.

Source: This button selects the track or bus the incoming audio is coming from.

Sidechaining and “Ducking” with Dynamics

The Dynamics module allows you to set up “sidechain” compression, which lets you temporarily lower the level of a target track or bus using the audible signal from a source track or bus.

For example, you may have a dialogue on a track or bus (the source) and another with music (the target), and you want the dialogue to be heard more prominently. This can be done by following the steps below:



Choosing the track header in the Fairlight timeline to add dynamics

- 1 Open the Dynamics plugin on the music track or bus, and switch on the Compressor.
- 2 Click the Source dropdown and select the dialogue track or bus.

TIP: Using a bus as the ducking source lets you use multiple sources to trigger changes in your target track.

- 3 Click the On button, which turns red, indicating that the Sidechain section is active.
- 4 While playing a section of your mix, set the amount of gain reduction by lowering the Threshold and raising the Ratio until the target audio is attenuated to a level that makes the dialogue more audible.
- 5 If needed, adjust the Hold parameter to ensure the gain reduction doesn't fluctuate too wildly.
- 6 You may also need to adjust the Release parameter to stop the gain reduction from ending too abruptly.

Clicking the Listen button lets you hear the incoming signal if you need to confirm that the correct input signal is coming into the Limiter. This button turns yellow when active.

Limiter

The third set of dynamics parameters let you apply limiting, which imposes a targeted maximum level allowed for a signal. Limiting is similar to compression but essentially has a very high ratio.

- **Limiter:** Enables the Limiter.

Threshold: Sets the threshold above which limiting will occur. As the Limiter is designed to hard limit the output, this also essentially sets maximum allowable output level. The default is -21 dB. The range is from -50 to 0 dB. It's best to set the amount of limiting by listening, and looking at the amount of gain reduction in the meter. Too much limiting can "squash the life out of the sound" as all of the internal dynamics of a mix can be lost. Used properly, you can smooth out the final mix and achieve higher average level. You can also watch the yellow animated balls in the display to see how the real time limiting response is affected by the controls.

Attack: Adjusts the attack rate time constant of the sidechain detector. The default is 0.71 ms. The range is 0 to 100 ms.

Hold: Once the input level falls below the threshold, Hold keeps limiting active until the hold time has elapsed, when release time will start. Default is 0 ms. The range is from 0 to 4000 ms.

Release: Adjusts how quickly the sidechain detector stops applying limiting when the signal falls below the threshold. The default is 90 ms. The range is 50 to 4000 ms.

NOTE: When limiting is applied to the final output of your channel (or to an entire mix), you often will need to use the Makeup Gain slider to maximize the output level (if a master) or to match the original level of an individual track (unity). You can audition the difference in level by bypassing the Dynamics module overall, or the Limiter only.

EQ

Each Mixer channel strip has a 6-band equalizer EQ, with a mini-EQ graph that displays the current EQ curve, bypasses the plugin, and opens it when double-clicked.

This plugin offers a choice of high and low shelving, parametric or notch response curves, and high and low pass filters with selectable slopes up to 24 dB/octave, with controls for boosting and attenuating different frequency ranges.

Each band has controls for the filter type (Bell, Lo-Shelf, Hi-Shelf, Notch), Frequency, Gain, and Q-factor (sharpness of the band), with the available controls for each band of EQ changing depending on the filter type.

You also have a choice of four response curves, from the Native Fairlight response curve to emulations of classic EQs.



The channel strip EQ window



The channel strip's EQ Indicator.
(Left) EQ is adjusted, and (Right) EQ is flat (no EQ).

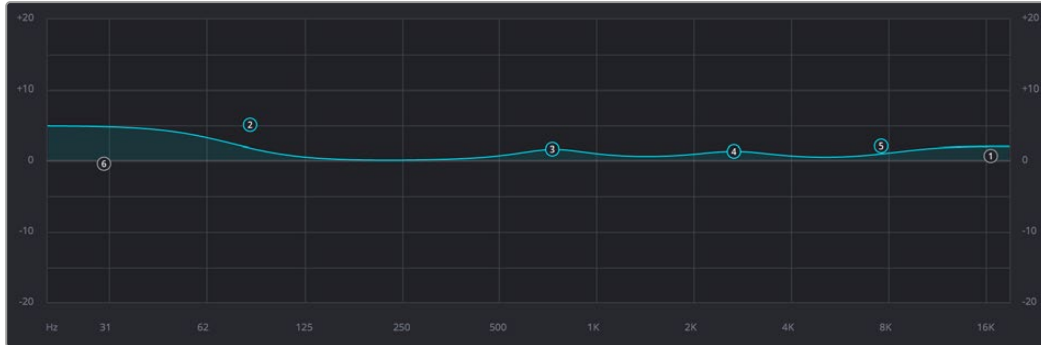
Master EQ Controls

The Equalizer window has the following overall controls:

- **Enable button:** Turns the plugin on and off.
- **Reset button:** The circular Reset icon on the upper right of the EQ resets all parameters to their default settings and values.
- **Equalizer Type:** This dropdown offers the Native Fairlight response curve and three emulations of classic mixing console EQs.
 - Earth:** Native Fairlight response (Default)
 - Air:** SSL 4K
 - Ice:** Neve V
 - Fire:** Focusrite
- **Preset Menu:** Your EQ settings can be saved as presets by clicking the Plus button to the left of the Preset menu.
- **Gain fader:** This post-EQ fader lets you boost or attenuate the processed signal to compensate for your EQ adjustments.

The EQ Graph

The EQ graph provides a visual representation of the current EQ settings, with numbered “handles” for each active EQ band, which can be dragged in any direction to change the corresponding frequency and level. You can also use your mouse wheel to adjust the Q (bandwidth) of each band.



The EQ graph

Dragging the numbered handles on this graph in turn modifies the parameters of the corresponding band, and conversely, changing each band’s parameters will also alter the EQ graph, which serves the additional purpose of providing a graphical representation of the equalization curve being applied to that track. If you have a mouse wheel, it can be used to control Q.

Bands 1 and 6

The outer two sets of band parameters are for processing the lowest or highest frequencies of an audio signal.



Band 1 controls active

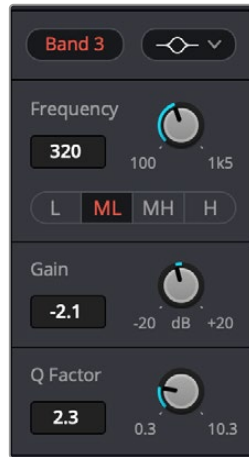


Band 6 controls active

- **Band filter type:** Clicking the Band 1 or Band 6 buttons activates or deactivates their respective controls.
- **Band filter type:** Bands 1 and 6 are switched off by default, but when active, they each allow access to filtering options below:
 - Band 1:** Low Shelf, Bell, High Shelf, and High Pass.
 - Band 6:** Low Pass, Low Shelf, Bell, and High Shelf.
- **Frequency:** Adjusts the center frequency of the EQ band.
- **Filter Slope:** The high and low pass filters offer selectable slopes starting at 6dB/octave up to 24dB/octave.
- **Gain:** When bands 1 and 6 are set to a bell or shelving response curve, this parameter cuts or boosts the selected frequency with a range of ± 20 dB.

Bands 2 – 5

The middle four sets of band controls let you make a wide variety of equalization adjustments. They're on by default to make it easy to begin making adjustments.



The Band 3 controls

- **Band enable button:** Clicking these Band buttons activates or deactivates their respective controls.
- **Band filter type:** Bands 2 to 5 offer the following filtering options: Low Shelf, Peaking, Notch, and High Shelf.
- **Frequency:** Adjusts the center frequency of the EQ adjustment.
- **L, ML, MH, H Buttons:** These buttons restrict the center frequency of a band to one of the following frequency ranges: Low, Medium Low, Medium High, or High.
- **Gain:** This control is available when a band is set to a bell or shelving response curve and adjusts the level of the center frequency within a range of ± 20 dB.
- **Q Factor:** This parameter is available when a band is set to a bell response or when the Equalizer type is set to Fire and adjusts the bandwidth of an EQ band, with a range of .3 to 10.3. Smaller Q factors widen the bandwidth, while higher values make it narrower, determining how much neighboring frequencies are also affected as the center frequency is adjusted.

Bus Sends

Bus Sends let you route a channel strip's source signals to a bus destination, with control over level, pan, mute, and pre/post fader routing. You can create bus sends on channels that go to any bus destination you've created.

Some common mixing tasks that sends can be useful for:

- **Submixing:** For example, to render separate mixed outputs that are different formats, such as stereo, 5.1 surround, or Dolby Atmos, all at the same time.
- **Feeding signals to a shared effect:** For example, route to a reverb plugin, where post fader sends on various mixer channels feed a reverb placed on a bus channel strip by varying amounts.
- **Headphone cue mixes:** Create separate pre-fader mixes for the talent so you don't disturb what you're listening to on your main monitor speakers.

Creating Sends

To create a Bus Send and access its controls:

- Click on the assign button with the plus sign (+) in the Bus Sends area of the Mixer.
- A dropdown menu appears with a list of busses; select one to assign.
- Your send is now created.

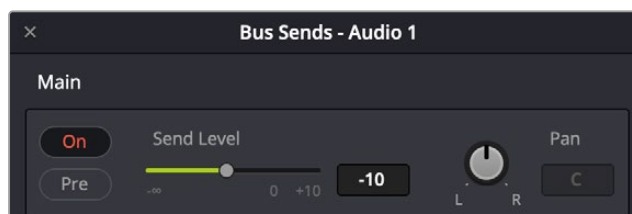
You can hover over the rectangular name button and access a disable/bypass button, an access icon to show the Bus Send window for the channel, and an “x” to delete the send. A tooltip appears showing the send bus name.



Hovering over the rectangular name button will show send controls.

Accessing the Bus Sends Window

You can access the Bus Sends window to control a channel's sends by clicking in the center of the name button.



Bus Send window controls

Each Bus Send you create exposes the following controls in the Bus Send window:

- **On:** Turns the send on or off.
- **Pre:** Switches the send to use pre- or post-fader routing. Enabling this button allows that track to send levels to the destination bus before level adjustments on that track are applied.
- **Send Level:** Adjusts the amount of signal sent from the selected feed to the send bus. The range is OFF to +10dB.
- **Pan:** Provides panning across Bus Send destinations.

Uses for Pre-fader Routing

One common use of pre-fader sends is for cue mixes to ADR or voice over talent, where the artist wants to hear a completely different mix than the control room mix. Using pre-fader sends and setting up a complete mix to a bus that feeds the headphone system allows a mix that is completely independent.

Pre-fader sends can also be used for special creative effects that might involve a dialogue channel being sent to a reverb plugin via a send. If the send is routed pre-fader, when the main channel fader is lowered to close to zero, the dialogue is still being fed to the reverb, so the dialogue will sound extremely reverberant (almost all of the signal is “wet”). Gradually raising the channel fader will make it sound like the voice is moving closer to the listener, changing the reverb balance and sounding more present.

Bus send levels are shown in the rectangles for each send and have the color codes assigned to the destination busses.



Bus Send levels shown on the Mixer channel strip

Legacy Bus Sends – “Auxiliaries”

Prior to DaVinci Resolve 17, the Bus Send panel was called Auxiliaries. Any projects created prior to DaVinci Resolve 17 will use the legacy Fixed Busing. The only difference between the two is the use of “Aux” to refer to sends in the windows and titles.

Pan

The Fairlight Mixer provides 2D and 3D pan controls that support various audio formats ranging from mono and stereo to conventional 5.1 and 7.1 surround, as well as advanced immersive formats such as Dolby Atmos, Ambisonics, Auro 3D, and NHK 22.2.

Channel Strip Pan Controls

The Pan section of each Channel Strip includes a Panning Indicator offering visualization and panning control of left, right, front, and back.

Although you can open the Panner to access additional parameters by double-clicking the Panning indicator, you can also do the following in the channel strip Pan section while the Panner is closed:

Click and drag within the Pan Indicator to adjust the basic panning position.

- Single click to bypass panning.
- Command-drag to adjust the width on multichannel tracks.
- View the Boom level for surround channels, which is displayed as a horizontal blue line at the bottom of the Panning Indicator.

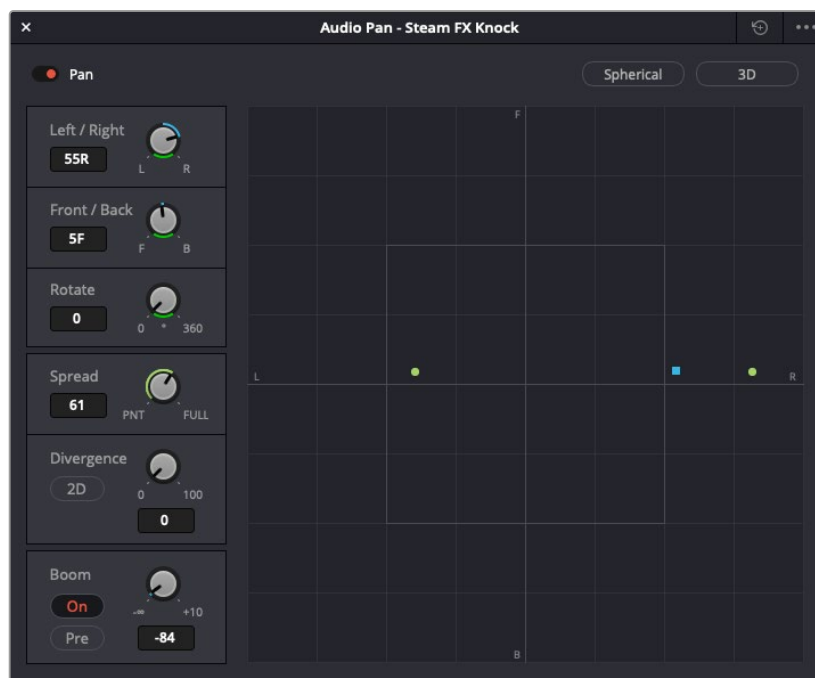


Channel strip
Pan Indicator for
a stereo track

2D Panner Controls

Default View

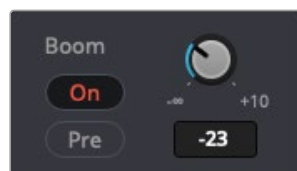
The Pan window's control set varies based on the mapping of the source audio. Panning adjustments can be made by dragging the icons in the Panner Viewer or using the controls on the left side of the window.



The Panner control window – Default 2D view

The default 2D version of the Audio Pan window includes the following controls:

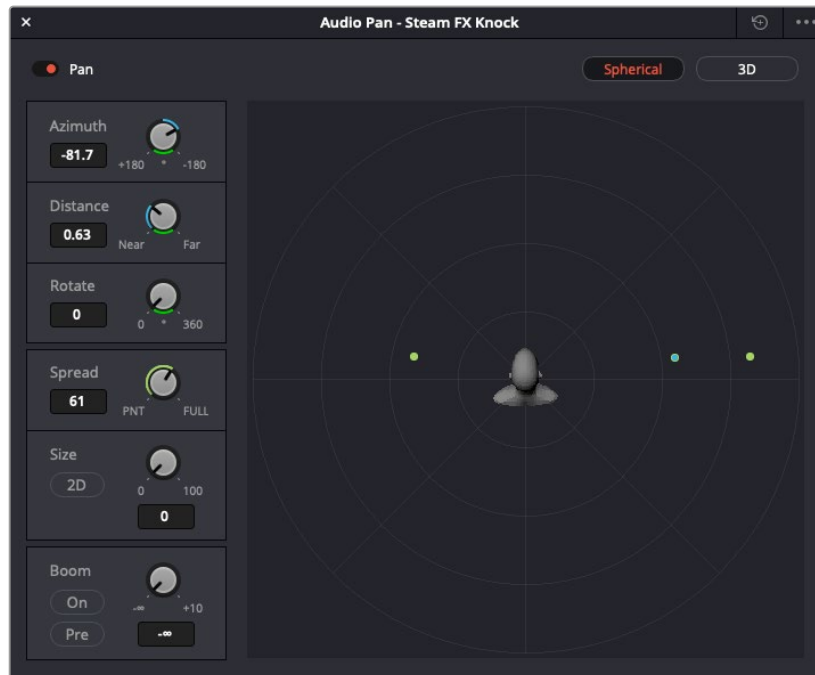
- **Pan Enable:** Switches the Panner on and off.
 - **Spherical:** This button changes the Panner interface from the default 2D view to the 2D Spherical top-down view.
 - **3D:** Clicking this button switches the interface to the 3D Cartesian view, offering controls for working with immersive audio formats such as Dolby Atmos, Ambisonics, Auro 3D, and NHK 22.2, as described in the 3D Pan Controls section below.
 - Clicking the Spherical and 3D buttons switches the interface to the 3D Spherical view.
 - **Left/Right:** Changes the balance of the signal between the left and right speakers, depending on the speaker configuration you're mixing for.
 - **Front/Back:** Lets you adjust the balance of the signal between the front and rear speakers based on the surround format you're mixing in.
 - **Rotate:** This controller rotates the sound source (or sources) around the listening position in the center of the sound field.
 - **Spread:** In the case of multichannel audio, this adjusts the spread or distance between the sound sources. This is measured on a numeric scale of 1-99, where 1 is labeled PNT ("Point") and 99 is labeled FULL.
 - **Divergence:** Determines the amount of signal bleed or spillover of a signal from one speaker into adjacent loudspeakers, allowing the sound to occupy more of the sound field, with less focus toward a single speaker position. The amount of signal bleed is represented by green horizontal lines emanating from either side of each speaker position that increases in length as you raise the parameter value.
- 2D:** Clicking this button changes the appearance of the divergence indicators to circles.
- **Boom:** This adjusts the send level to the LFE part of the mix (the Low-Frequency Extension subwoofer system).
- On:** Enables the Boom output.
- Pre:** Routes the LFE signal pre-fader.



Boom parameter

TIP: To constrain left/right position as you adjust panning, hold down the Shift key. You can reset any rotary control to its default value by double-clicking it.

Spherical View



The Panner control window – Spherical view

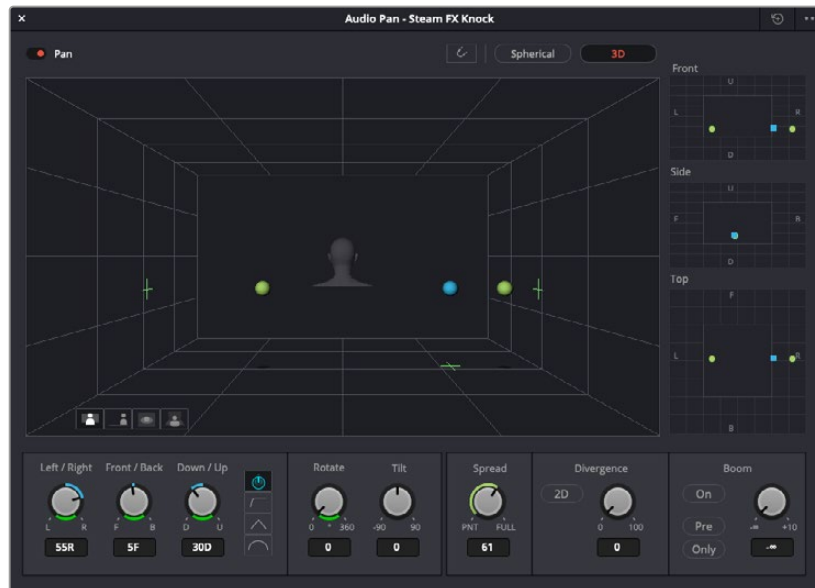
Clicking the Spherical button at the top of the 2D Panner changes the interface to Spherical view, which has the controls listed below including two parameters for Ambisonic panning:

- **Azimuth:** This control lets you rotate the sound source around the circumference of the sphere.
- **Distance:** This control adjusts the distance between the signal source and the listening position at the center of the sphere.
- **Rotate:** This controller rotates the sound source (or sources) around the listening position in the center of the sound field.
- **Spread:** In the case of multichannel audio, this adjusts the spread or distance between the sound sources. This is measured on a numeric scale of 1-99, where 1 is labeled PNT ("Point") and 99 is labeled FULL.
- **Size:** This parameter is not active at the time of writing.
- **2D:** This parameter is not active at the time of writing.
- **Boom:** This adjusts the send level to the LFE part of the mix (the Low-Frequency Extension subwoofer system).
- **On:** Enables the Boom output.
- **Pre:** Routes the LFE signal pre-fader.

3D Panner Controls

DaVinci Resolve includes 3D Cartesian and Spherical versions of this Panner, which lets you do spatial audio positioning when working in advanced surround formats such as Dolby Atmos, Ambisonics, Auro 3D, and NHK 22.2.

Cartesian View



The 3D Panner control window - Default Cartesian view

Clicking the 3D button changes the Panner interface to the default Cartesian view, which includes a few more controls than the 2D Pan window:

- **Pan enable:** Toggles the Panner on and off.
- **Snap:** This button toggles Speaker Snap mode on and off for a given object. During playback, Speaker Snap mode moves an audio object to the active speaker nearest its established location, to prevent issues like phantom panning.
- **Panner viewer:** A large 3D representation of the virtual room, with a blue sphere representing the position of the track's audio.
Perspective Buttons: These buttons let you view the room from different preset angles. You can freely rotate it by holding down Command-Option-Shift (Ctrl + Alt + Shift in Windows) while dragging it. If you want to return to a default view, click one of the perspective buttons.
- **Front panner:** A 2D panning control that lets you make adjustments along the horizontal Left/Right axis and the vertical Down/Up axis.
- **Side panner:** A 2D panning control that lets you make adjustments along the horizontal Front/Back axis and the vertical Down/Up axis.
- **Top panner:** A 2D panning control that represents the horizontal left/right axis the vertical front/back axis, letting you make these specific spatial adjustments.
- **Left/Right:** Changes the balance of the signal between the left and right speakers, depending on the speaker configuration you're mixing for.

- **Front/Back:** Lets you adjust the balance of the signal between the front and rear speakers based on the surround format you are mixing in.
- **Down/Up:** This lets you change the position of a sound source along the vertical axis.
- **Height Mode Presets:** Located to the right of the Down/Up control, these three height presets lock the interaction of the Front/Back and Down/Up parameters together, providing fixed panning paths that raise the height of a sound source to its maximum (100U), while moving it from one end of the sound field to the other.
 - Freeform:** This is the default setting for this section, which lets you adjust the Down/Up and Front/Back controls independently.
 - Flyover:** Moves the sound source upwards in a sharp arc to 100U, and then from one end of the sound field to the other while maintaining maximum height.
 - Triangular:** This preset moves the source diagonally up to 100U and immediately back down again as it travels from one end of the sound field to the other.
 - Arc:** Moves the sound source upwards in an arc to 100U and immediately back down in the same manner as it travels from one end of the sound field to the other.
- **Rotate:** This controller rotates the sound source or mix around the listening position in the center of the sound field.
- **Tilt:** This knob tilts a sound source or mix up to ninety degrees to either side so that it vertically rotates or “swings” around the listening position in the center of the sound field.
- **Spread:** In the case of multichannel audio, this adjusts the spread or distance between the sound sources. This is measured on a numeric scale of 1-99, where 1 is labeled PNT (“Point”) and 99 is labeled FULL.
- **Divergence:** Determines the amount of signal bleed or spillover of a signal from one speaker into adjacent loudspeakers, allowing the sound to occupy more of the sound field, with less focus toward a single speaker position. The amount of signal bleed is represented by green horizontal lines emanating from either side of each speaker position that increases in length as you raise the parameter value.
 - 2D:** Clicking this button changes the appearance of the divergence indicators to circles.
- **Boom:** This adjusts the send level to the LFE part of the mix (the Low-Frequency Extension subwoofer system).
 - On:** Enables the Boom output.
 - Pre:** Routes the LFE signal pre-fader.
 - Only:** Allows the panner to output an LFE signal only from the channel. The Boom level line in the channel strip Pan Indicator is gray when this button is active.

Cartesian View — Dolby Atmos



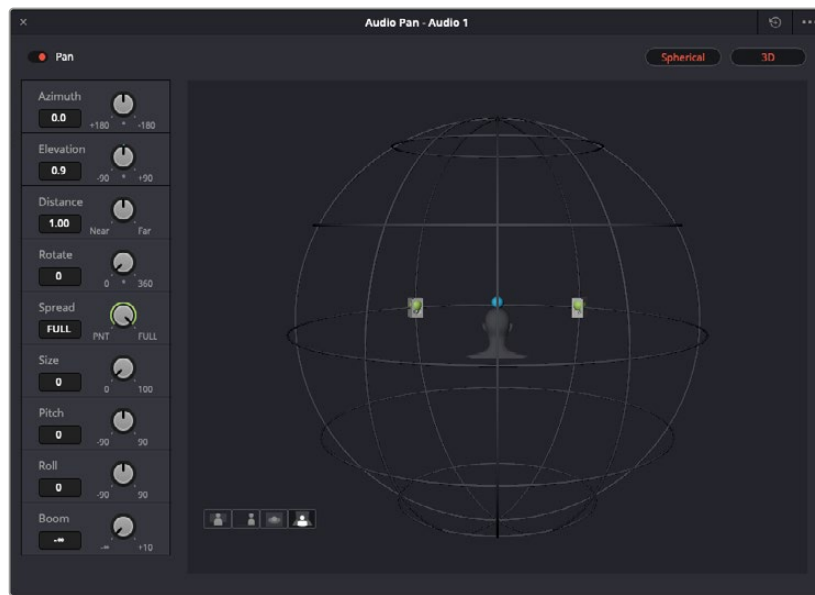
The 3D Panner control window - Cartesian view – Dolby Atmos

When working in a Dolby Atmos workflow, the 3D Cartesian view of the Panner includes all the features mentioned above for the default Cartesian view, with the addition of the following controls located at the top of the interface:

NOTE: These controls are not available on Dolby Atmos busses.

- **Elevation Button:** This button hides and reveals the four top speakers in the Viewer.
- **Zone:** In addition to the option to view all speakers this dropdown menu offers five others that let you include or exclude certain speakers from this object's mix:
 - No Back:** Hides the rear surround speakers.
 - No Sides:** Hides the side surround speakers.
 - Center Back:** Shows only the Center and Rear speakers.
 - Screen Only:** Shows only the Center and Left and Right Front speakers.
 - Surr (Surround) Only:** Show only the Surround speakers.

Spherical View



The 3D Spherical Panner control window

The 3D Spherical Panner lets you position your source anywhere in the sound field, including above and below the listener. Spherical panning is based on Azimuth, Elevation, and Distance.

This view of the 3D panner offers the following controls:

- **Azimuth:** This control rotates the sound source around the circumference of the sphere.
- **Elevation:** This raises and lowers the sound source towards the top and bottom of the sphere, which allows you to place it above or below the listener if you want.
- **Distance:** This control adjusts the distance between the signal source and the listening position at the center of the sphere.
- **Rotate:** Rotates the sound source (or sources) around the listening position in the center of the sound field.
- **Spread:** This control adjusts the spread or distance between the sound sources. This is measured on a numeric scale of 1-99, where 1 is labeled PNT (“Point”) and 99 is labeled FULL.
- **Size:** This parameter is not active at the time of writing.
- **Room:** This parameter is not active at the time of writing.
- **Pitch:** This knob tilts the sound source or mix, forward or backward while rotating or “swinging” it up to ninety degrees around the center listening position.
- **Roll:** This knob tilts a sound source or mix, up to ninety degrees to either side so that it vertically rotates or “swings” around the center listening position.
- **Panner viewer:** A large 3D representation of the spherical sound field, with a blue sphere representing the position of the track’s audio.
- **Perspective Buttons:** These buttons let you view the Sphere from different preset angles. You can freely rotate it by holding down Command-Option-Shift (Ctrl + Alt + Shift in Windows) while dragging it. If you want to return to a default view, click one of the perspective buttons.

Bus Assignment Buttons

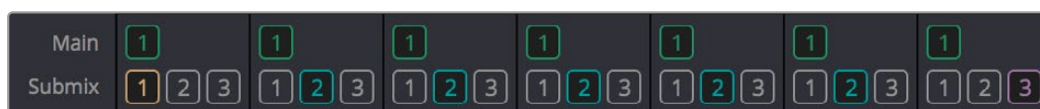
To route a bus created in the Bus Assignment dialog directly on the Mixer, click the + sign to see a dropdown menu of available busses. To route multiple busses simultaneously, Command-Click (Ctrl+Click) each of your choices in the dropdown menu.



The Bus Outputs panel is where you can route bus signals.

Legacy Mixer Bus Assignment

In the Legacy Mixer the bus assignments have two sets of buttons that let you route audio from one channel strip's output to Sub and Main busses that you've set up for your mix.



The channel strip's Bus Assignment buttons

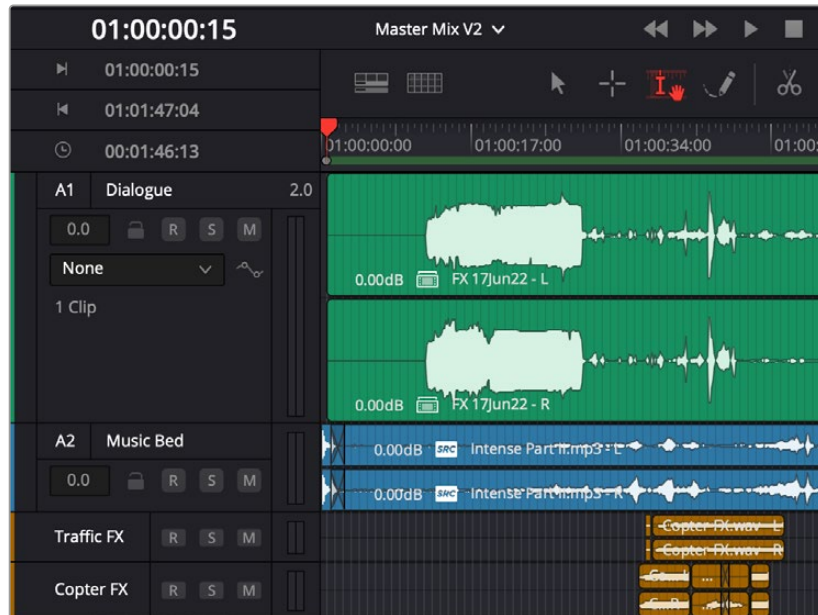
- **Main:** These buttons let you assign a track or sub's channels to one or more of the main busses.
- **Submix:** These buttons let you assign that track's channels to one or more submix busses.

Nested Audio Timelines

Edited timelines can be combined with other timelines by dragging the desired nested timeline onto an empty track in the destination timeline. This is a very powerful and useful feature.

For general information about nested timelines, see *Chapter 43, "Take Selectors, Compound Clips, and Nested Timelines."*

For example, you might have created a master timeline that will contain your master mix, and you'd like to bring the work done on a separate timeline that is focused on dialogue editing. The dialogue timeline could now appear as a single nested timeline, represented as single track in the master mixing timeline.

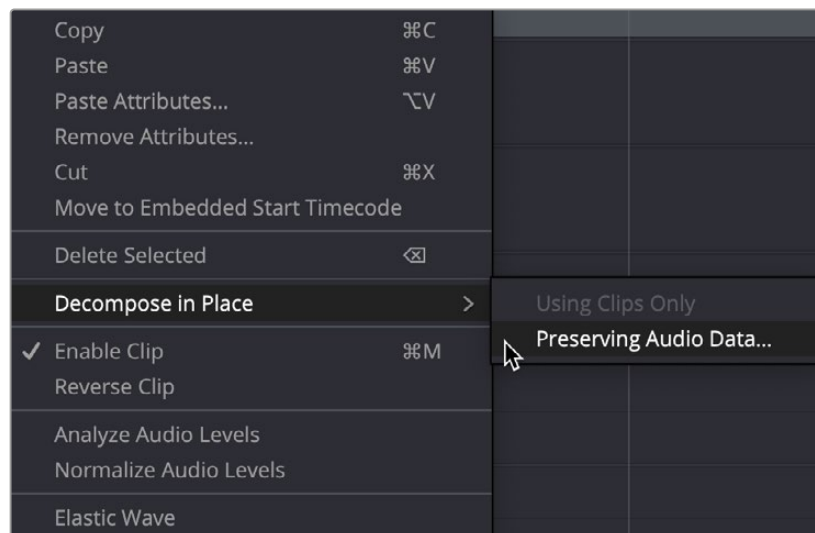


Track A1, Dialogue, is a stereo representation of an edited timeline.

If you want to work with the individual tracks and clips again, you can use a process called decompose to change that single nested timeline track back to the edited sequence. There are a number of options to decompose to the original set of tracks, with several bussing options available.

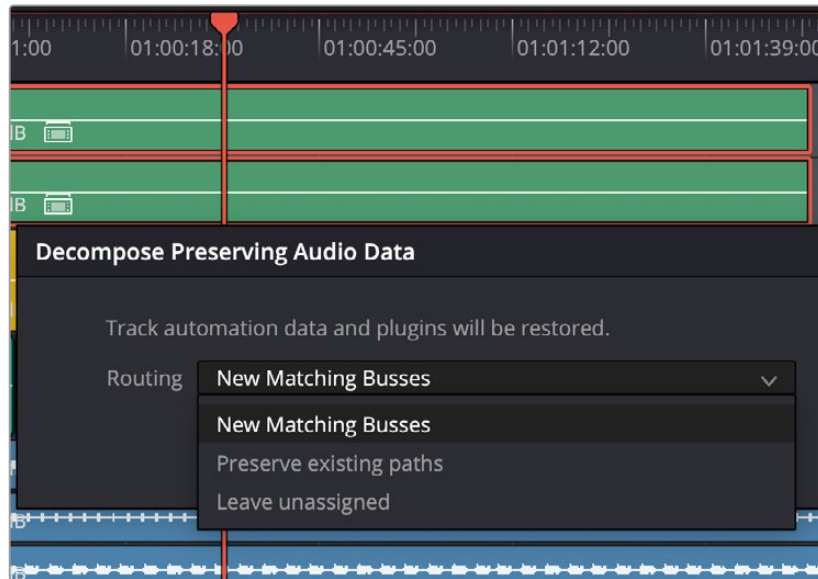
This image shows what was a separate, edited timeline of tracks of dialogue and is now a single track, A1 Dialogue, brought into the destination master mix timeline. By right-clicking the clip, a contextual menu allows you to decompose the track in place, back into its original tracks and clip edits.

There are two decompose options for this. On the Fairlight page, you can Preserve Audio Data, and when on the Edit page, you can Preserve Audio Data as well as decompose by Using Clips Only.



Decompose in Place contextual menu

Once you've chosen to decompose the track into its prior Timeline elements, then the Decompose Preserving Audio Data dialog box appears, making several more options are available.



Options for signal routing when decomposing a clip

There are three options when choosing to decompose a nested timeline in the Fairlight page. All three options preserve mix and plugin automation on the source tracks; they only differ in terms of routing. The choices are:

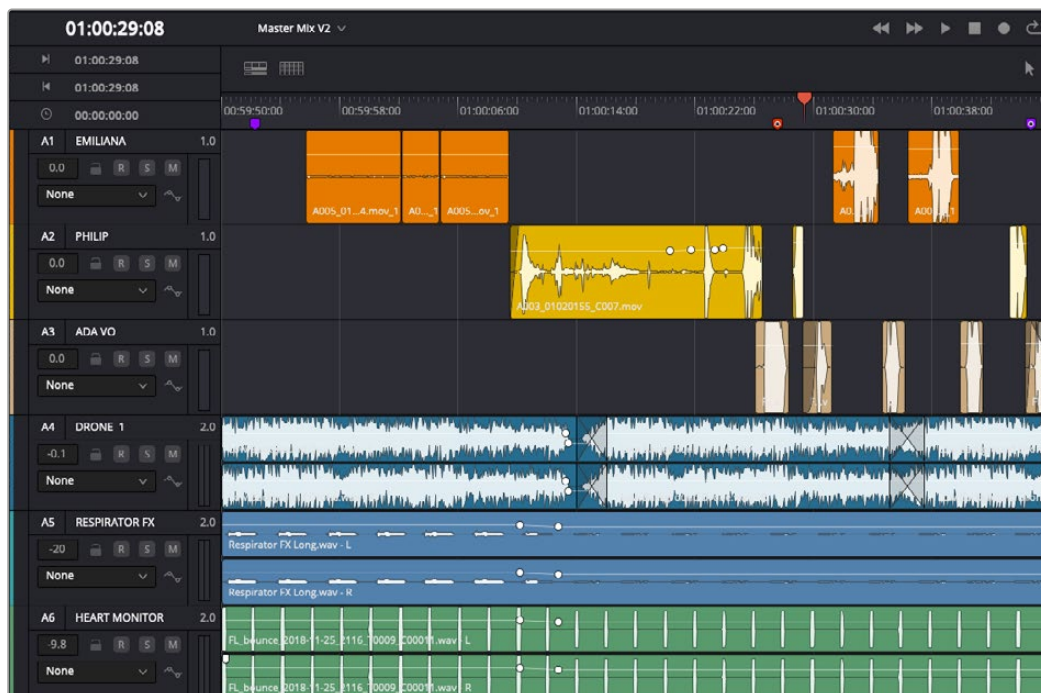
New Matching Busses: All of the busses from the nested source timeline are brought into the current timeline as separate, new busses. These busses will have all of the processing, routings, settings, and automation carried over from the nested source timeline. Any preexisting bus routing in the destination timeline is left unaltered.

This option is best when you want combine different mix layouts and to preserve all preexisting bus routing in the destination timeline, while adding the bus topology from the nested source timeline to the destination to create a single, larger entity.

Preserve Existing Paths: This option preserves all of the master bus information in the current timeline and routes matching busses from the decomposed nested timeline to preexisting, identical busses in the destination timeline.

This option is best when the nested source timeline and the destination timeline share the same bus routing structure, as might be the case of a common layout template. In this case, you just want to add the tracks to the destination timeline and route them to the same designations in the master. You still get all of your mix automation and plugins carried across as well.

Leave Unassigned: This option leaves the new tracks unassigned when created, ready to be routed in whatever way you need.



Decomposed version of the nested dialogue timeline showing the original dialogue tracks and edits

Nested audio timelines can be very useful when managing large projects with many tracks and sophisticated routing. Multiple editors can be working on various aspects at one time. For instance, a dialogue editor could be doing the dialogue and ADR in one timeline, an effects editor can be designing and spotting effects in another timeline, a music editor can be cutting and spotting music in another timeline, all separately. Then these interim timelines can be imported later to a master timeline for a final main mix.

This is also useful for picture editors who don't want to have to deal with many tracks of audio when cutting picture. When working in the Edit page, when a track sequence is decomposed, it can also revert back to the original track layout with all of the audio clips ready to be further edited.

VCA Faders

A VCA fader is used to control the level of multiple tracks with a single control. You can assign multiple faders to a dedicated VCA, and a VCA channel strip then appears at the right of the Mixer. VCAs let you simultaneously adjust the underlying level of multiple faders using one VCA master fader and can be helpful in managing the levels of complex collections of audio tracks.



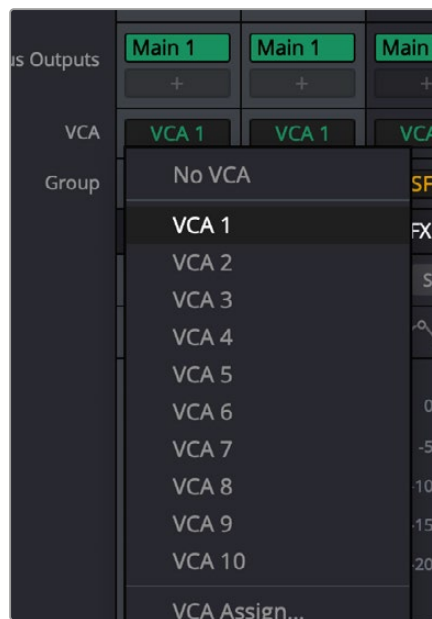
Group Labels in each control strip, above the control strip label

The controls for doing this are found in the VCA and Group label area of each mixer channel strip, which shows which VCA or Track Group each fader has been assigned to.

Making Fader VCA Assignments

You can assign any channel strip to one of 64 VCAs by right-clicking the VCA label area and choosing a VCA from the dropdown that appears. If a fader is already assigned to a VCA, you can choose “No VCA” to remove it.

NOTE: To reduce unneeded clutter, the VCA Assignment dropdown menu will only display 10 VCAs initially; others are added as needed. If the first 10 have all been assigned, the menu display will add VCAs from the available pool, one a a time.



Right-clicking a VCA Label reveals a dropdown menu that lets you assign faders to or remove faders from that VCA.

TIP: You can quickly assign VCAs for a selected group of channel strips, or to all channel strips, by holding down the Option key (Mac) or the Alt key (Windows) for all selected tracks or Command-Option (Mac) or Control-Alt (Windows) for all mixer channel strips prior to performing the operation. These shortcuts can save a lot of time in your workflow.

Using VCA Faders

Once you’ve assigned multiple faders to a VCA, a dedicated channel strip for that VCA appears at the right side of the Mixer. Making adjustments to the VCA channel strip simultaneously controls all the faders, solo buttons, and mute buttons of all channel strips that are members of that group, as seen below.



Adjusting the VCA 1 fader also adjusts the SYNC, VO, and Drum Hit faders that are assigned to that VCA.

When controlling faders belonging to a VCA, you can still move each individual channel strip fader independently to make relative adjustments.

- Channel strip faders only move together when you adjust the VCA fader.
- While the VCA fader is being moved, each individual fader's relative offset from other faders controlled by the VCA is maintained.

For example, as seen in the screenshot above, the VCA 1 fader is moving the faders in tracks SYNC, VO, and Drum Hit, while each individual channel strip fader in this group maintains its offset from the others.

Recording Fader Automation for VCAs

You can record automation data to a VCA fader, and all faders in that VCA group will follow even though they're not actually automated themselves. This makes it easy to record complex automation involving multiple faders when you still might want the freedom to finely adjust each individual fader later on.

Nesting VCAs

VCAs can also be "nested," where one or more VCAs can be controlled by yet another VCA. To nest a VCA, simply assign the VCAs you want have controlled to yet another "master" VCA.

Arm, Solo, and Mute Buttons

These controls are identical to the controls found on the track header of each timeline audio track.

- **Record Arm:** Enables a track for recording when the Record button on the transport is engaged (highlights red when enabled).
- **Solo:** Lets you mute all other tracks in order to play a track you need to focus on in isolation (highlights green when enabled).

If Solo is enabled for multiple tracks, all soloed tracks will play, and all non-soloed tracks will be muted.

Solo status affects rendering, so if one or more tracks are soloed, the muted tracks won't be output or rendered.

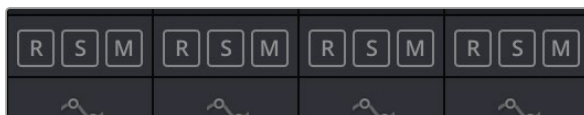
- **Solo Safe:** Command-Shift-click a Solo button to put it into Solo Safe mode (highlights blue when enabled).
Channel strips set to Solo Safe always play even if Solo is enabled for other tracks.
Solo Safe can be useful for effect returns busses with effects like reverb where you may want to hear the reverb when tracks feeding it are soloed.
- **Exclusive Solo:** When active, clicking the Solo button on a track takes other soloed tracks out of Solo mode. This means only one track can be soloed and heard at any given time, which is useful if you want to compare audio sources, to hear sonic differences when comparing levels or processing.

NOTE: Soloing a track in Exclusive Solo mode will not affect a track that is in Solo Safe mode. If you want to include that track, you need to manually deactivate the Safe Solo button first.

Disabled by default, Exclusive Solo can be enabled by selecting it in the Fairlight menu. You can also map a key command to this function in the Keyboard Customization dialog.

For more information on Keyboard Customization, see *Chapter 4, "System and User Preferences."*

- **Mute:** Turning on Mute disables audio playback from that track (highlights orange when enabled). This affects rendering, so if one or more tracks are muted, they won't be output or rendered.



The channel strip's Record Arm, Solo, and Mute buttons

TIP: You can click a button and drag across multiple channel strips to turn that button on or off for multiple tracks easily.

Fader Controls

Each track's vertical fader lets you control the level that's output by that track, either by using your mouse, or using a physical fader of your Fairlight console or third party control surface. If you're working with a console, then the onscreen faders serve as a visual reference of what levels are set.

dB Indicator: Numerical at the top of the track indicates the volume, in decibels, that track is currently set to.

Fader: Each track's vertical fader can be dragged with your mouse or other pointing device to adjust the volume of that track and perform automation recording. Dragging up increases volume, dragging down decreases volume. Fader handles turn red while you record level automation, and they turn green when automation has been recorded for that track.

Methods of adjusting channel faders:

- **To change level:** Click and drag any fader up or down.
- **To reset the level to default level of 0 dB:** Double-click a fader's handle. This does not work after you've recorded automation for a track, unless you erase the automation first.



A Fader in the Fairlight Mixer

Metering Options

DaVinci Resolve has a variety of metering options that allow you to tailor metering to your workflow. The settings appear in Project Preferences > Fairlight in the Audio Metering section, but they affect mixer, Fairlight effects, or master metering on the Cut and Edit pages.

Level Metering Options

You can choose the response characteristic of the level meters for channel strips and Fairlight FX.

The Meter Type dropdown allows you to choose between IEC 60268-18, Digital VU, and Custom response characteristics. Both meter types have separate “hold and fall” metering, allowing you to see the highest peak that is reached.

IEC 60268-18: A digital PPM-type meter with reference standard of -18 dBFS, a fast response to peaks, and a slower release characteristic. This is the default in DaVinci Resolve and is used in all Blackmagic Design software and hardware products.

Digital VU: A dual-value meter, showing the peak level as a single segment with fast ballistics, and the RMS (volume unit) as a bar graph. It has a far faster quasi linear decay characteristic, making it easy to monitor average levels while allowing sounds with quick transients and decays to be more easily tracked. This option can be best for audio editing and mixing.

Custom: Allows each aspect of the meter response to be chosen, including level detector, scale, peak, peak, and peak indication.

Pre-fader Metering

By default, “Pre fader metering on tracks” is unchecked (disabled), setting the metering on channel strips to be post (after) the fader level and mute button. For example, if the source audio file has peaks that are hitting -2 dBFS, and the channel fader is lowered from 0 to -10, then the peaks that show on the meter will now be at -12.

When pre-fader metering is on, the metering point is set to be before the fader, and is not affected by the fader’s position. However, the meter is affected by clip gain settings or key framing as these level changes occur before the clip’s signal enters the mixer’s signal path.

With pre-fader metering, you could lower the fader to be fully off (minus infinity) and the meter will still read the level of the source signal. This allows you to always be aware of the source signal and what it is doing before being altered by mixer’s controls or processing.

Many video editors prefer to work with post-fader metering, but most audio mixers prefer to work with pre-fader in order to be aware of what the actual source level is at any time.

Solo and Pre-fader Metering

When pre-fader metering is enabled, soloing one or more tracks will show a lighter “ghost” version of meter activity on non-soloed tracks, allowing you to always see audio levels on tracks that are muted by the soloing process. This makes it easy to focus on tracks that are actually audible while still seeing activity on those that are not.

If pre-fader metering is off (the default), when one or more tracks are soloed, no metering occurs on tracks unsoloed tracks.



Pre-fader metering with light shaded meters on non-soloed tracks

Target Loudness Level

Allows the target loudness level in LUFS for the master Loudness meters to be set to your desired output target level. For example, DaVinci Resolve's default loudness standard target is -23 LUFS, but the YouTube target LUFS specification is -14 LUFS. If -14 LUFS is set, the "0" mark on the loudness meter scale moves to this level, allowing you to focus your master mix levels towards that loudness standard.

Bouncing Audio

Bouncing audio refers to mixing and rendering audio from one or more Timeline tracks onto another track of the Timeline, in the process "baking in" processor intensive effects and complicated or intricate audio edits to create a new continuous piece of audio media. The bounced file is written to the directory location specified by the Project Settings > Capture and Playback panel > "Save clips to" field.

There are two commands available for bouncing audio on the Fairlight page:

- Timeline > Audio > Bounce Selected Tracks in Place
- Timeline > Audio > Bounce Mix to Track

To use Bounce Selected Tracks to New Layer:

- 1 Set In and Out points to define the range of the Timeline you want to bounce. If you don't do this, nothing will happen.
- 2 Command-click the track headers or mixer channel strips of all tracks you want to bounce in order to select them.
- 3 Choose Timeline > Audio > Bounce Selected Tracks to New Layer.

The audio on each track is processed and rendered and appears as the top layer of audio on that track. While View > Show Audio Track Layers is turned off, it will appear as if the new bounced audio is the only clip on that track. However, the original audio (with any unrendered clip-based effects) is still available as the bottom of the stack of layered audio on that track; turning on View > Show Audio Track Layers will reveal this.

The bounced audio is a new audio media file that's written to the directory location specified by the Project Settings > Capture and Playback panel > "Save clips to" field.

To use Bounce Mix to Track:

- 1 Choose Timeline > Bounce Mix to Track. The Bounce Mix to Track window appears, showing all busses that are currently available.
- 2 In the Destination Track column, set which mixes you want to bounce by choosing either New Track, or choosing a specific existing track from the dropdown menus.
- 3 Click OK.

The specified mix is processed, mixed, and bounced to the specified track as a new piece of audio. This creates new audio media that's written to the directory location specified by the Project Settings > Capture and Playback panel > "Save clips to" field.

TIP: There's also a Bounce Audio Effects command in the contextual menu of audio clips in the Timeline that have clip-based audio effects applied to them. For more information, see Chapter 180, "Audio Effects."

For more information on exporting clips, ranges, and files, see Chapter 175, "Editing Basics in the Fairlight Page."

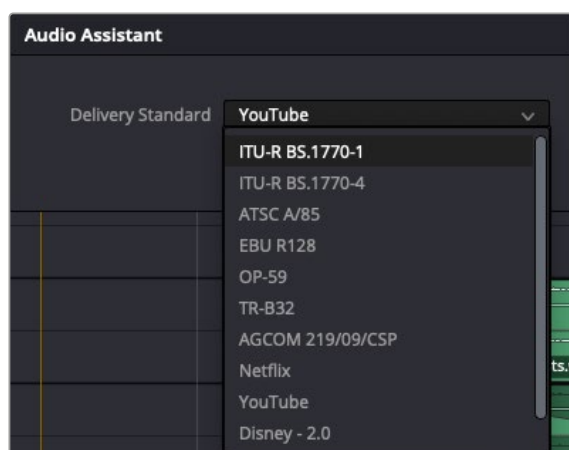
AI Audio Assistant (Studio Version Only)

With the AI Audio Assistant, DaVinci Resolve can automatically organize and color-code your tracks, even out your dialogue levels, and adjust the Fairlight mixer faders to create a professional-quality mix of your music, sound effects, and dialogue.

During this process, your dialogue will also be cleaned up, and if needed, Voice Isolation and De-Essing will be applied. Ducking will be applied to music tracks, any required automation will be written, and the appropriate mastering and finalizing plugins will be added and adjusted to help achieve the correct loudness level for your chosen delivery standard.

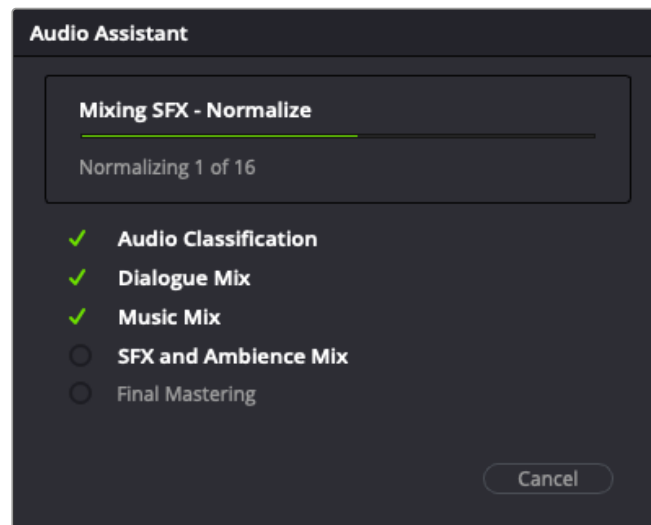
To use AI Audio Assistant:

- 1 Open a timeline containing unmixed audio. The initial fader levels really don't matter as Audio Assistant will handle all fader adjustments during the mixing process.
- 2 To begin the automatic mixing process, choose Timeline > AI Tools > Audio Assistant from the Timeline menu, select the Delivery Standard for your mix from the drop-down menu, and click the Auto Mix button.



Audio Assistant dialog - Delivery Standard

- 3 The Audio Assistant dialog changes to display the progress of the mixing process with an option to cancel it.



Audio Assistant dialog - Mix Progress

- 4 When the mixing process has concluded, you can play back the mix and, if needed, make further adjustments or completely undo it.

NOTE: If a track is in the wrong category, right-click the corresponding track header and choose the correct category from the Track Category section of the contextual menu, and then rerun the Audio Assistant.

Third-Party Control Panel Support for Mixing

DaVinci Resolve supports HUI- and MCU-compatible third-party mixing control panels with up to eight faders, such as the Mackie MCU Pro Control Surface, connected via USB MIDI, and selectable in the Control Panels panel of the Resolve System Preferences. Supported basic panel controls that correspond to Fairlight features at the time of this writing include the following.

Transport controls including:

- Rewind (REW)
- Fast Forward (FF)
- Stop (double-press the Stop button for Home in the Timeline)
- Play (double-press the Play button to "play again")
- Record (press Record+Play to begin recording if one or more tracks is record-enabled)
- Jog control (Press SCRUB and rotate the jog wheel)

Channel Strip controls including:

- Rotary controls for panning, with Rotary Value display and Rotary touch
- Alphanumeric Track Name display
- Record button to record-enable tracks (only works if an input is patched to a track input)
- Solo, with a double-press selecting or de-selecting that track
- Mute
- Select to select the track corresponding to that channel strip
- Fader control and optional Level display
- Channel and Fader Bank buttons to move left and right among banks of channel strips
- Double-press the Fader Bank Left button to move the playhead Home in the mixer (track 1)
- Double-press the Fader Bank Right to move to the Master channel strips

Marker buttons:

- Marker to add a marker
- Pressing the Marker+Stop buttons sets Home
- Pressing the Marker+FF/REW buttons jumps the playhead forward or backward

Additional supported controls for control panels that have them include:

- Master Solo clear/restore
- Undo (pressing the Undo+Option buttons does Redo)
- Arrows to move the selection
- Zoom horizontal and vertical controls
- Audio Tracks, to turn Automation On/Off
- Write/Trim/Touch/Latch switches
- Nudge controls
- Cut, Copy, and Paste

Monitoring controls include:

- Level control
- Dim
- Mute
- Alt Speaker

For more information on third-party mixing control panel support, see the Blackmagic Support site at www.blackmagicdesign.com/support/family/davinci-resolve-and-fusion.